

Assignment 2: TCP Connection Performance Analysis using Mininet and Wireshark

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2. Mininet Topology

```
graph TD
    s1 --- h1
    s1 --- h2
```

h1 = TCP Server (10.0.0.1)
h2 = TCP Client (10.0.0.2)
s1 = OpenFlow Switch

3. Explanation of TCP Client-Server Communication

Ans : TCP (Transmission Control Protocol) provides reliable communication between a client and a server.

Working Process

1. The client initiates a connection with the server.
2. A TCP three-way handshake establishes the connection.
3. The client sends messages in the format: MSG_ID|MESSAGE_SIZE|MESSAGE_DATA

Example: 3|512|AAAA...

4. The server receives the message and extracts: Message ID, Message Size, Message Data
5. The server replies with: ACK|MSG_ID|RECEIVED_SIZE

Example: ACK|3|512

6. The client measures response time and throughput.
7. After all transmissions are completed, the connection is terminated.

4. Difference Between Persistent and New-Connection Modes

Feature	Persistent Mode	New-Connection Mode
Connection Establishment	One time	For every message
Number of Handshakes	One	Multiple
Connection Closing	One time	After every message
Overhead	Low	High
Response Time	Lower	Higher
Efficiency	Better	Less efficient

Resource Usage

Less

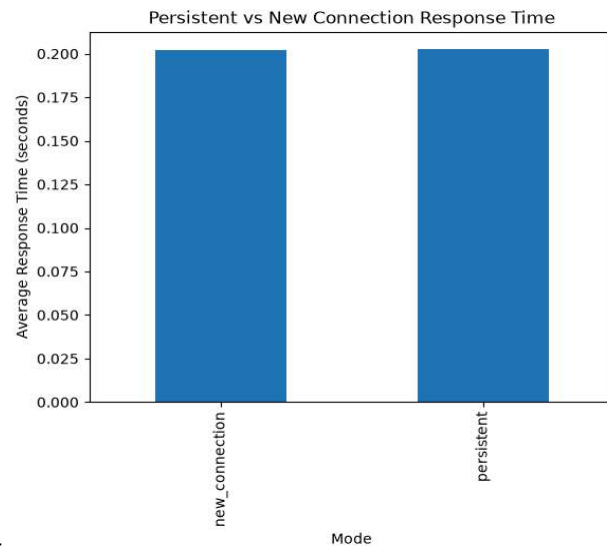
More

5. Result Table

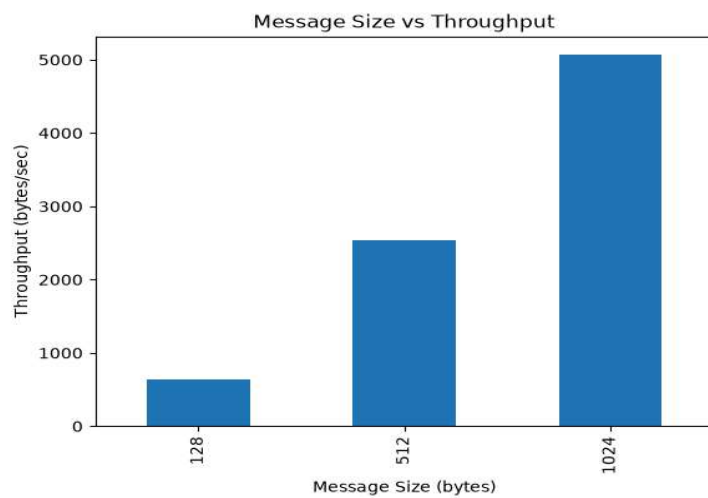
roll_no	name	mode	message_size	total_messages	average_response_time	throughput_bytes_per_sec
CSB24051	Piyush Kumar	persistent	128	10	0.20261	631.7557
CSB24051	Piyush Kumar	persistent	512	10	0.20209	2533.531
CSB24051	Piyush Kumar	persistent	1024	10	0.202049	5068.076
CSB24051	Piyush Kumar	new_connection	128	10	0.20244	632.2859
CSB24051	Piyush Kumar	new_connection	512	10	0.202206	2532.076
CSB24051	Piyush Kumar	new_connection	1024	10	0.202398	5059.328

6. All three graphs

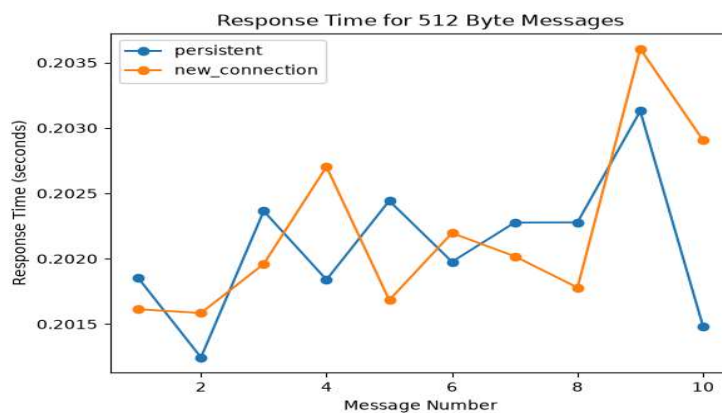
1. mode_vs_response_time.png



2. message_size_vs_throughput.png



3. message_response_time.png



7. Wireshark screenshots.

1. persistent_handshake.png

No.	Time	Source	Destination	Protocol	Length	Info
8	63.094590675	10.0.0.2	10.0.0.1	TCP	74	43102 → 5000 [SYN] Seq=0 Win=42340 Len=0 MSS=1460 SACK_PERM TSval=1086947863 TSecr=0 WS=512
9	63.145315196	10.0.0.1	10.0.0.2	TCP	74	5000 → 43102 [SYN, ACK] Seq=0 Ack=1 Win=43440 Len=0 MSS=1460 SACK_PERM TSval=3314067011 TSecr=1086947863 WS=512
10	63.297599599	10.0.0.2	10.0.0.1	TCP	66	43102 → 5000 [ACK] Seq=1 Ack=1 Win=42496 Len=0 TSval=1086948274 TSecr=3314067011

2. Persistent_data_packets.png

11	63.299064580	10.0.0.2	10.0.0.1	TCP	200	43102 → 5000 [PSH, ACK] Seq=1 Ack=1 Win=42496 Len=134 TSval=1086948275 TSecr=3314067011
12	63.349765246	10.0.0.1	10.0.0.2	TCP	66	5000 → 43102 [ACK] Seq=1 Ack=135 Win=43520 Len=0 TSval=3314067215 TSecr=1086948275
13	63.349805012	10.0.0.1	10.0.0.2	TCP	75	5000 → 43102 [PSH, ACK] Seq=1 Ack=135 Win=43520 Len=9 TSval=3314067216 TSecr=1086948275
14	63.501365562	10.0.0.2	10.0.0.1	TCP	66	43102 → 5000 [ACK] Seq=135 Ack=10 Win=42496 Len=0 TSval=1086948478 TSecr=3314067216
15	63.501991871	10.0.0.2	10.0.0.1	TCP	200	43102 → 5000 [PSH, ACK] Seq=135 Ack=10 Win=42496 Len=134 TSval=1086948479 TSecr=3314067216
16	63.552338716	10.0.0.1	10.0.0.2	TCP	75	5000 → 43102 [PSH, ACK] Seq=10 Ack=269 Win=43520 Len=9 TSval=3314067418 TSecr=1086948479
17	63.704806415	10.0.0.2	10.0.0.1	TCP	200	43102 → 5000 [PSH, ACK] Seq=269 Ack=19 Win=42496 Len=134 TSval=1086948681 TSecr=3314067418
18	63.756054495	10.0.0.1	10.0.0.2	TCP	75	5000 → 43102 [PSH, ACK] Seq=19 Ack=403 Win=43520 Len=9 TSval=3314067621 TSecr=1086948681
19	63.907046811	10.0.0.2	10.0.0.1	TCP	200	43102 → 5000 [PSH, ACK] Seq=403 Ack=28 Win=42496 Len=134 TSval=1086948884 TSecr=3314067621
20	63.957919071	10.0.0.1	10.0.0.2	TCP	75	5000 → 43102 [PSH, ACK] Seq=28 Ack=537 Win=43520 Len=9 TSval=3314067824 TSecr=1086948884
21	64.110577674	10.0.0.2	10.0.0.1	TCP	200	43102 → 5000 [PSH, ACK] Seq=537 Ack=37 Win=42496 Len=134 TSval=1086949087 TSecr=3314067824
22	64.161570337	10.0.0.1	10.0.0.2	TCP	75	5000 → 43102 [PSH, ACK] Seq=37 Ack=671 Win=43520 Len=9 TSval=3314068027 TSecr=1086949087
23	64.313917649	10.0.0.2	10.0.0.1	TCP	200	43102 → 5000 [PSH, ACK] Seq=671 Ack=46 Win=42496 Len=134 TSval=1086949290 TSecr=3314068027
24	64.365163608	10.0.0.1	10.0.0.2	TCP	75	5000 → 43102 [PSH, ACK] Seq=46 Ack=805 Win=43520 Len=9 TSval=3314068231 TSecr=1086949290
25	64.516410699	10.0.0.2	10.0.0.1	TCP	200	43102 → 5000 [PSH, ACK] Seq=805 Ack=55 Win=42496 Len=134 TSval=1086949493 TSecr=3314068231
26	64.567413372	10.0.0.1	10.0.0.2	TCP	75	5000 → 43102 [PSH, ACK] Seq=55 Ack=939 Win=43520 Len=9 TSval=3314068433 TSecr=1086949493
27	64.719872822	10.0.0.2	10.0.0.1	TCP	200	43102 → 5000 [PSH, ACK] Seq=939 Ack=64 Win=42496 Len=134 TSval=1086949696 TSecr=3314068433
28	64.770971194	10.0.0.1	10.0.0.2	TCP	75	5000 → 43102 [PSH, ACK] Seq=64 Ack=1073 Win=43520 Len=9 TSval=3314068637 TSecr=1086949696
29	64.923222688	10.0.0.2	10.0.0.1	TCP	200	43102 → 5000 [PSH, ACK] Seq=1073 Ack=73 Win=42496 Len=134 TSval=1086949900 TSecr=3314068637
30	64.974126162	10.0.0.1	10.0.0.2	TCP	75	5000 → 43102 [PSH, ACK] Seq=73 Ack=1207 Win=43520 Len=9 TSval=3314068840 TSecr=1086949900
31	65.126325390	10.0.0.2	10.0.0.1	TCP	201	43102 → 5000 [PSH, ACK] Seq=1207 Ack=82 Win=42496 Len=135 TSval=1086950103 TSecr=3314068840
32	65.176927145	10.0.0.1	10.0.0.2	TCP	76	5000 → 43102 [PSH, ACK] Seq=82 Ack=1342 Win=43520 Len=10 TSval=3314069043 TSecr=1086950103

3. Persistent_connection_close.png

33	65.328380490	10.0.0.2	10.0.0.1	TCP	66	43102 → 5000 [FIN, ACK] Seq=1342 Ack=92 Win=42496 Len=0 TSval=1086950305 TSecr=3314069043
34	65.328329400	10.0.0.2	10.0.0.1	TCP	74	43110 → 5000 [SYN] Seq=0 Win=42340 Len=0 MSS=1460 SACK_PERM TSval=2766163231 TSecr=0 WS=512
35	65.379246376	10.0.0.1	10.0.0.2	TCP	74	5000 → 43110 [SYN, ACK] Seq=0 Ack=1 Win=43440 Len=0 MSS=1460 SACK_PERM TSval=606967591 TSecr=2766163231 WS=512
36	65.379283673	10.0.0.1	10.0.0.2	TCP	66	5000 → 43102 [FIN, ACK] Seq=92 Ack=1343 Win=43520 Len=0 TSval=3314069245 TSecr=1086950305

4. new_connection_multiple_handshakes.png

342	79.957969241	10.0.0.1	10.0.0.2	TCP	74	5000 → 59960 [SYN, ACK] Seq=0 Ack=1 Win=43440 Len=0 MSS=1460 SACK_PERM TSval=1089274435 TSecr=1154411726 WS=512
343	80.108420316	10.0.0.2	10.0.0.1	TCP	66	59950 → 5000 [ACK] Seq=1033 Ack=12 Win=42496 Len=0 TSval=2071647366 TSecr=1593089146
344	80.108925610	10.0.0.2	10.0.0.1	TCP	66	59960 → 5000 [ACK] Seq=1 Ack=1 Win=42496 Len=0 TSval=1154411927 TSecr=1089274435
345	80.108930133	10.0.0.2	10.0.0.1	TCP	1097	59960 → 5000 [PSH, ACK] Seq=1 Ack=1 Win=42496 Len=1031 TSval=1154411928 TSecr=1089274435
346	80.159572233	10.0.0.1	10.0.0.2	TCP	66	5000 → 59960 [ACK] Seq=1 Ack=1032 Win=44032 Len=0 TSval=1089274636 TSecr=1154411928
347	80.160652415	10.0.0.1	10.0.0.2	TCP	76	5000 → 59960 [PSH, ACK] Seq=1 Ack=1032 Win=44032 Len=10 TSval=1089274637 TSecr=1154411928
348	80.312256791	10.0.0.2	10.0.0.1	TCP	66	59960 → 5000 [ACK] Seq=1032 Ack=11 Win=42496 Len=0 TSval=1154412130 TSecr=1089274637
349	80.312283658	10.0.0.2	10.0.0.1	TCP	66	59960 → 5000 [FIN, ACK] Seq=1032 Ack=11 Win=42496 Len=0 TSval=1154412131 TSecr=1089274637
350	80.312285918	10.0.0.2	10.0.0.1	TCP	74	59964 → 5000 [SYN] Seq=0 Win=42340 Len=0 MSS=1460 SACK_PERM TSval=2893759143 TSecr=0 WS=512
351	80.362898366	10.0.0.1	10.0.0.2	TCP	74	5000 → 59964 [SYN, ACK] Seq=0 Ack=1 Win=43440 Len=0 MSS=1460 SACK_PERM TSval=3440875073 TSecr=2893759143 WS=512
352	80.362941448	10.0.0.1	10.0.0.2	TCP	66	5000 → 59964 [FIN, ACK] Seq=11 Ack=1033 Win=44032 Len=0 TSval=1089274840 TSecr=1154412131
353	80.514525415	10.0.0.2	10.0.0.1	TCP	66	59964 → 5000 [ACK] Seq=1 Ack=1 Win=42496 Len=0 TSval=2893759345 TSecr=3440875073
354	80.514549405	10.0.0.2	10.0.0.1	TCP	66	59960 → 5000 [ACK] Seq=1033 Ack=12 Win=42496 Len=0 TSval=1154412333 TSecr=1089274840
355	80.514551573	10.0.0.2	10.0.0.1	TCP	1097	59964 → 5000 [PSH, ACK] Seq=1 Ack=1 Win=42496 Len=1031 TSval=2893759345 TSecr=3440875073
356	80.565338776	10.0.0.1	10.0.0.2	TCP	66	5000 → 59964 [ACK] Seq=1 Ack=1032 Win=44032 Len=0 TSval=3440875275 TSecr=2893759345
357	80.565553389	10.0.0.1	10.0.0.2	TCP	76	5000 → 59964 [PSH, ACK] Seq=1 Ack=1032 Win=44032 Len=10 TSval=3440875276 TSecr=2893759345
358	80.716956984	10.0.0.2	10.0.0.1	TCP	66	59964 → 5000 [ACK] Seq=1032 Ack=11 Win=42496 Len=0 TSval=2893759547 TSecr=3440875276
359	80.716981871	10.0.0.2	10.0.0.1	TCP	66	59964 → 5000 [FIN, ACK] Seq=1032 Ack=11 Win=42496 Len=0 TSval=2893759548 TSecr=3440875276
360	80.717531318	10.0.0.2	10.0.0.1	TCP	74	59970 → 5000 [SYN] Seq=0 Win=42340 Len=0 MSS=1460 SACK_PERM TSval=13706897 TSecr=0 WS=512
361	80.767671430	10.0.0.1	10.0.0.2	TCP	66	5000 → 59964 [FIN, ACK] Seq=11 Ack=1033 Win=44032 Len=0 TSval=3440875477 TSecr=2893759548
362	80.767712697	10.0.0.1	10.0.0.2	TCP	74	5000 → 59970 [SYN, ACK] Seq=0 Ack=1 Win=43440 Len=0 MSS=1460 SACK_PERM TSval=3694313852 TSecr=13706897 WS=512

8. Answers to Questions

Q1. Why does TCP use a three-way handshake?

TCP uses a three-way handshake to establish a reliable connection between the client and server. It synchronizes sequence numbers and ensures that both devices are ready to exchange data.

Q2. Which mode had lower response time?

Persistent mode had lower response time because only one TCP connection was established and reused for all messages.

Q3. Why does new-connection mode have more overhead?

In new-connection mode, each message requires:

- Connection establishment (three-way handshake)
- Data transmission
- Connection termination

These repeated operations increase delay and consume additional resources, resulting in higher overhead.

Q4. Which mode is better for CampusChat and why?

Persistent mode is better for CampusChat because users continuously exchange messages. Maintaining a single connection reduces delay and improves communication speed.

Q5. Which mode is suitable for NetAttend and why?

New-connection mode is suitable for NetAttend because attendance information is sent occasionally. Opening a new connection for each request is simple and sufficient for such infrequent communication.

Q6. How does message size affect throughput?

As message size increases, throughput generally increases because a larger amount of useful data is transmitted while the connection overhead remains nearly the same. Thus, larger packets utilize network resources more efficiently.

Q7. How did Wireshark help verify TCP behavior?

Wireshark helped by:

- Capturing TCP packets.
- Displaying SYN, SYN-ACK, and ACK packets.
- Verifying connection establishment and termination.
- Showing that persistent mode uses one connection while new-connection mode repeatedly creates connections.
- Confirming the correctness of packet transmission and TCP operation.